

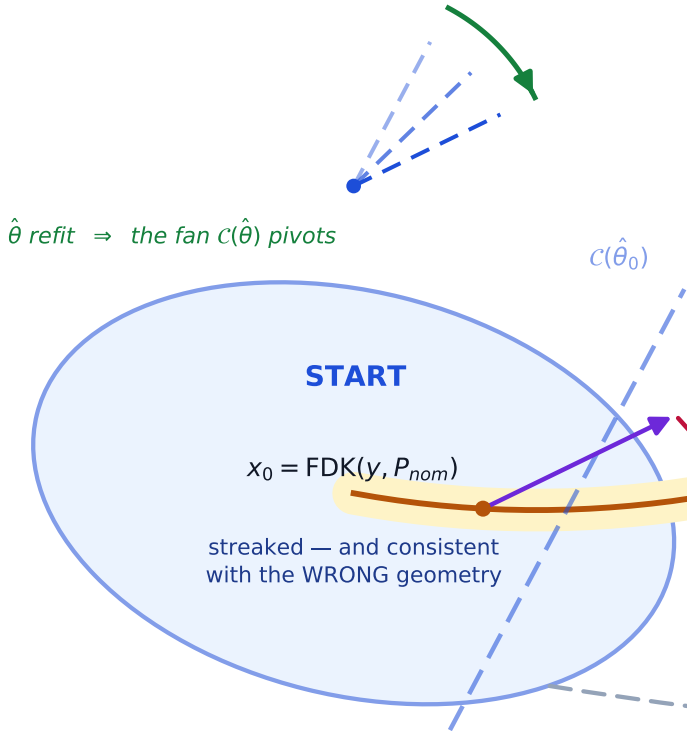
Why the loop works: a learned path to the clean manifold, held on the physics at every step

the prior supplies the DIRECTION (it was trained on this very path); the measured data supplies the CONSTRAINT that keeps the direction honest

ONE STEP OF THE LOOP (x50)

-  PREDICT — the prior steps toward $\mathcal{M}_{clean}$
it knows the direction: it was TRAINED on this path
-  ESTIMATE — $\hat{\theta}$ is refit on that better image.
It moves θ , not x : it RE-AIMS $c(\hat{\theta})$ — the pivot below.
-  CORRECT — the data step pulls it back onto $c(\hat{\theta})$; the TV corrector keeps it regular

prior alone:
a plausible image that does NOT
explain the measurement



GEOMETRY BRIDGE $\{\text{FDK}(y, P_{nom}T(s\theta))\}$

every point on it is a REAL reconstruction of the REAL data —
and it is exactly the path the prior was trained on ($v_\psi \approx dx_t/dt$)

landing: $\bar{\theta}$ and the carried x_t

the ESTIMATE stage is what re-aims it: $c(\hat{\theta}_0) \rightarrow c(\hat{\theta}_k) \rightarrow c(\theta_{true})$,
until the feasible set finally CONTAINS the clean image.

$c(\hat{\theta}) = \{x : A_{\hat{\theta}}x \approx y\}$ — the constraint that PHYSICALLY EXISTS:
the measured projections. It is not a regularizer we chose, it is the scan.

data step alone, at the geometry we started with ($\hat{\theta} = 0$): perfectly consistent — and still streaked

the three moves alternate $N = 50$ times — no single one reaches $\mathcal{M}_{clean}$